MIL-D-48865 (PA)
30 March 1976

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, PA500 PARTS FOR, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by Picatinny Arsenal, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, loading, assembling and packing for one type of electric detonator designated as Detonator, Electric, PA500.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packing
MIL-G-45204	-	Gold Plating, Electrodeposited
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

FEDERAL

TT-I-1795	-	Ink, Marking Stencil, Opaque for Porous Surfaces

STANDARDS

MILITARY

MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-650	-	Explosives: Sampling Inspection and Testing

FSC: 1375

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DRAWINGS

PICATINNY ARSENAL

9297952 - Detonator, Electric, PA500
8796522 - Marking Diagram and Sealing for Wood Packing Boxes

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (Lastest Revisions)".)

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and Assemblies.-The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9297952 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Moisture Content of Lead Azide.-The moisture content of the lead azide used in loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.2.

3.4 Moisture Content of HMX.-The moisture content of the HMX used in loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.3.

3.5 Resistance of Shorting Clip.-The resistance of the shorting clip shall not be greater than one tenth of the resistance of the bridge wire.

3.6 Functioning

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3.6.1 Output.-The detonators shall fire on the first attempt, and shall make a minimum (min.) depression of 0.005 inches in the steel witness plate when tested as specified in 4.5.8.1.

3.6.2 Waterproofness.-The detonator shall comply with the requirements of 3.6.1 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours, at a temperature of 75 degrees F. plus or minus 5 degrees, when tested as specified in 4.5.8.2.

3.6.3 Time.-The detonator shall fire within 200 micro-seconds when tested as specified in 4.5.8.3.

3.7 Environmental

3.7.1 Resistance of Lead Circuit, Thermal Shock and Vibration.-The detonators shall comply with the resistance requirement of the lead circuit on dwg. 9297952 and with the requirements of 3.6.1 after being exposed to thermal shock and vibration and tested as specified in 4.5.9.1.

3.7.2 Cold Temperature.-The detonator assembly shall comply with the requirements of 3.6.1 after being conditioned at -65 ± 5°F for two hours and tested as specified in 4.5.9.2.

3.7.3 Hot Temperature.-The detonator assembly shall comply with the requirements of 3.6.1 after being conditioned at + 165 ± 5°F for two hours and tested as specified in 4.5.9.3.

3.8 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

3.9 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility For Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Cup	9297962	50
Cup, Loading Assembly	9297961	50
Electrode	9297959	50
Tubing, Glass	9297957	50
Eyelet	9297956	50
Plug Assembly	9297955	50
Bridged Plug Assembly	9297954	50
Ignition Plug Assembly	9297953	50
Detonator,Electric PA500	9297952	325

4.3.2 Inspections to be Performed.-See MIL-A-48078, and Table I specified herein.

4.3.3 Rejection.-(See MIL-A-48078).

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	PA500 Electric Detonator and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup (Dwg. 9297962) Examination for Defects	50	-	3.2	4.4.2.1
	Eyelet (Dwg. 9297956) Examination for Defects	50	-	3.2	4.4.2.2
	Tubing, Glass (Dwg. 9297957) Examination for Defects	50	-	3.2	4.4.2.3
	Electrode (Dwg. 9297959) Examination for Defects	50	-	3.2	4.4.2.4
	Plug Assembly (Dwg. 9297955) Examination for Defects	50	-	3.2	4.4.2.5
	Gold Plating of Electrodes	50	-	3.2	4.5.4
	Pull Test of Plug Assembly	50	-	3.2	4.5.5
	Bridged Plug Assembly (Dwg. 9297954) Examination for Defects	50	-	3.2	4.4.2.6

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48865 (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	PA500 Electric Detonator and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ignition Plug Assembly (Dwg. 9297953)				
	Examination for Defects	50	-	3.2	4.4.2.7
	Cup, Loading Assembly (prior to loading upper charge) (Dwg. 9297961)				
	Examination for Defects	50	-	3.2	4.4.2.9
	Cup, Loading Assembly (Dwg. 9297961)				
	Examination for Defects	50	-	3.2	4.4.2.10
	Detonator, Electric, PA500 (Dwg. 9297952)				
	Examination for Defects	(a)	-	3.2	4.4.2.11
	Insulation Test	(b)	-	3.2	4.5.6
	Resistance of Shorting Clip	(b)	-	3.5	4.5.7
	Functioning-Output	200	-	3.6.1	4.5.8.1
	Functioning-Waterproofness	50	-	3.6.2	4.5.8.2
	Functioning-Time	(c)	-	3.6.3	4.5.8.3
	Environmental Resistance of Lead Circuit	25	-	3.7.1	4.5.9.1
	Environmental Cold Temperature	25	-	3.7.2	4.5.9.2
	Environmental Hot Temperature	25	-	3.7.3	4.5.9.3

NOTES:

(a) The sample shall consist of the 325 detonators to be functioned.
(b) The sample shall consist of the 325 detonators examined in (a).
(c) The sample shall consist of those same detonators as the Functioning Output Test.

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer, in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant. The second lead azide lot shall not vary from the first lot by more than three serial lot numbers.

c. HMX from not more than one lot.

4.4.2 Examination.-(See MIL-A-48078).

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9297962
4.4.2.1	Cup				NEXT HIGHER ASSEMBLY 9297961
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of bottom		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Concentricity of inside diameter to outside diameter		0.40%	3.2	Gage
104	Contour of bottom radius		0.40%	3.2	Gage
105	Inside diameter		0.40%	3.2	Gage
106	Length		0.40%	3.2	Gage
107	Depth of spherical radius in bottom		0.40%	3.2	Gage
108	Dent or distortion		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48865 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Eyelet	1 OF 1	9297956
			NEXT HIGHER ASSEMBLY 9297955

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Diameter of holes		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
104	Location of holes		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48865 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9297957
4.4.2.3	Tubing, Glass				NEXT HIGHER ASSEMBLY 9297955
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Electrode		9297959
			NEXT HIGHER ASSEMBLY 9297955

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Straightness overall length		0.40%	3.2	Gage
102	Length		0.40%	3.2	Gage
103	Diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48865 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Plug Assembly		9297955
			NEXT HIGHER ASSEMBLY 9297954

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Non-sealing length		0.40%	3.2	Gage
102	Thickness of eyelet assembly		0.40%	3.2	Gage
103	Glass not flush or below flush		0.40%	3.2	Gage
104	Extension of glass on electrode improper		0.40%	3.2	Gage
105	Gold Plating of Electrodes		(a)	3.2	4.5.4
106	Pull Test of Plug Assembly	125		3.2	4.5.5
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

(a) Test plan per MIL-G-45204

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Bridged Plug Assembly	1 OF 1	9297954
			NEXT HIGHER ASSEMBLY 9297953

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cut length of electrodes		0.40%	3.2	Gage
102	Resistance between electrodes before assembly of wire, bridge		0.40%	3.2	Test
103	Resistance of wire, bridge between bridge posts after welding (a)		0.40%	3.2	Test
104	Bridge wire insecure		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
(a) Measured with a current limited to one milliampere.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Ignition Plug Assembly		9297953
			NEXT HIGHER ASSEMBLY 9297952

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of cracks, voids or separations in spot charge after drying		0.40%	3.2	Visual
102	Bridge wire visible through spot charge		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Cup, Loading Assembly (Prior to Loading Base Charge)		9297961
			NEXT HIGHER ASSEMBLY 9297952

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture Content of Lead Azide	(a)		3.3	4.5.2
102	Moisture Content of HMX	(a)		3.4	4.5.3
Minor	None defined				
NOTES:	(a) The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one (1) sample of each eight (8) hours of detonator production. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all detonators loaded with the nonconforming material, shall be rejected. The tests shall be performed as specified in 4.5.2 or 4.5.3 as applicable.				

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Cup, Loading Assembly (Prior to Loading Upper Charge)				9297961
					NEXT HIGHER ASSEMBLY 9297952
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height of base charge		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Cup, Loading Assembly		9297961
			NEXT HIGHER ASSEMBLY 9297952

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height of combined charges		0.40%	3.2	Gage
102	Assembly height (max.)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE DEFECTS

CLASSIFICATION OF DEFECTS & TESTS MIL-D-48865 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Detonator, Electric, PA500				9297952
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Cup cracked or split		100.0%	3.2	Visual
Major					
101	Resistance of lead circuit before shorting electrodes	(a)	100.0%	3.2	Test
102	Electrodes not shorted		100.0%	3.2	Visual
103	Insulation Test (with shorting clip)		100.0%	3.2	4.5.6
104	Length of electrodes		0.40%	3.2	Gage
105	Distance, crimped end to base		0.40%	3.2	Gage
106	Diameter of cup		0.40%	3.2	Gage
107	Depth of spherical radius in bottom (min)		0.40%	3.2	Gage
108	Crimp not full 360 degrees		0.40%	3.2	Gage
109	Resistance of shorting clip		0.40%	3.2	4.5.7
110	Functioning-output	200	-	3.6.1	4.5.8.1
111	Functioning-waterproofness	50	-	3.6.2	4.5.8.2
112	Functioning-time	(b)	-	3.6.3	4.5.8.3
113	Environmental resistance of lead circuit	25	-	3.7.1	4.5.9.1
114	Environmental cold temperature	25	-	3.7.2	4.5.9.2
115	Environmental hot temperature	25	-	3.7.3	4.5.9.3
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
(a) Measured with a current limited to one milliampere
(b) The sample shall consist of those same detonators at the same time as the functioning output test.

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section of MIL-A-48078 and 6.2 herein.

4.5 Test Method and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Moisture Content of Lead Azide.-The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample in sufficient quantity to perform the test shall be selected and tested as specified herein. The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Method 101.5 except, use a two gram sample and cover the dish with a sheet of polytetrafluoroethylene.

4.5.3 HMX.-The moisture content of HMX shall be determined as specified in 4.5.2.

4.5.4 Gold Plating of Electrodes.-The gold plating shall be verified as specified in MIL-G-45204. (Test samples shall not be returned to the lot).

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4.5.5 Pull Test of Plug Assembly.-The required (minimum) force shall be applied, for the specified time (45 ± 5 seconds), to the longer end of the electrodes on one end and to the eyelet on the other end, along the longitudinal axis of the eyelet, in a manner to pull the assembly apart. The assembly shall be classed defective if separation of assembly or any visible movement of the electrodes in the eyelet is observed. If one or more defectives occur the lot shall be rejected. (Test samples shall not be returned to the lot).

4.5.6 Insulation Test.-The required D.C. voltage (315 ± 10 volts) shall be applied between the shorted electrodes and the cup of all detonators. The resistance shall be a minimum of 22 megohms when measured between the electrodes and the cup. If any detonator fails to meet the requirements, it shall be classed defective. All defectives shall be removed from the lot.

4.5.7 Resistance of Shorting Clip.-The resistance of the detonator with shorting clip shall be measured between electrodes. The shorting clip shall then be removed and the resistance again measured. The resistance of the remaining assembly shall be at least eleven times as large as the resistance of the detonator with shorting clip. If any detonator fails to meet the requirements, it shall be classed defective. If one or more defectives occur the lot shall be rejected. The resistance should be measured with a current limited to one milliampere.

4.5.8 Functioning.-The detonator assemblies shall be temperature conditioned at 70 ± 10 °F for two (2) hours minimum prior to performing the functioning tests. The tests shall be performed as specified in MIL-STD-331, Test 301.1 except that the witness plate be 1/8 inch minimum and optionally round, 1.22 - .01 dia. Witness plate hardness shall be Rockwell B70 to B95. The base of the detonator shall be placed directly against the center of the witness plate and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to each electrode. The measurement of the depression in the witness plate shall be accomplished with a dial indicator equipped with a probe having a .003 ± .001 spherical radius.

4.5.8.1 Output.-Two hundred (200) detonators from each lot shall be subjected to this test. The detonators shall be assembled into a test fixture and fired by applying the energy from a 4.7 maximum microfarad capacitor charged to 20 volts maximum. The depression in the witness block shall be measured

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to determine compliance with the requirements. Functioning time shall be recorded. If any detonator fails to meet the requirements on the first attempt it shall be classed defective. If one or more defectives occur the lot shall be rejected.

4.5.8.2 Waterproofness.-Fifty (50) detonators from each lot shall be subjected to this test. The detonators shall be immersed in water at a two (2) to three (3) inch depth for 48 hours minimum. The required water temperature shall be maintained for the immersion period. At the end of the immersion time, the detonators shall be removed from the water, assembled into a test fixture and fired by applying the energy from a 4.7 maximum microfarad capacitor charged to 20 volts maximum. The depression in the witness block shall be measured to determine compliance with the requirements. Functioning times shall be recorded. If any detonator fails to meet the requirements on the first attempt it shall be classed defective. If one or more defectives occur the lot shall be rejected.

4.5.8.3 Time.-The functioning times of the two hundred (200) detonators subjected to the test in 4.5.8.1 shall demonstrate no time in excess of 250 microseconds (200-0-1) nor more than two (2) times greater than 200 microseconds (200-2-3). The lot shall be rejected if either or both requirements are not met.

4.5.9 Environmental Tests

4.5.9.1 Resistance of Lead Circuit, Thermal Shock and Vibration.-Twenty-five (25) detonators from each lot shall be subjected to this test. The detonators shall be tested in the following sequence:

a. Measure resistance between electrodes, using a current limited to one milliampere. The resistance shall be no less than 2.5 ohms nor more than 4.0 ohms.

b. Condition for two (2) hours minimum at -65 ± 5°F.

c. Remove from cold chamber and immediately transfer to the hot chamber.

d. Condition for two (2) hours minimum at +165 ± 5°F.

e. Remove from hot chamber and immediately transfer to the vibrator.

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f. Vibrate for 15 minutes minimum in a sweep covering the frequency range of 25-500-25 cycles per second parallel to the transverse orthagonal axis with base charge down and a 2 ± 0.2 gs peak.

g. Measure resistance as in a.

h. The detonators shall be functioned as specified in 4.5.8.1 and meet the requirements of 3.6.1.

If any detonator fails to meet the requirement on the first attempt it shall be classed defective. If one or more defectives occur, the lot shall be rejected.

4.5.9.2 Cold Temperature.-Twenty-five (25) detonators from each lot shall be subjected to this test. The detonators shall be conditioned at -65 ± 5°F for two (2) hours minimum. The detonators, maintained at the conditioning temperature, shall be functioned as specified in 4.5.8.1. If any detonator fails to meet the requirements on the first attempt it shall be classed defective. If one or more defectives occur the lot shall be rejected.

4.5.9.3 Hot Temperature.-Twenty-five (25) detonators from each lot shall be subjected to this test. The detonators shall be conditioned to +165 ± 5°F for two (2) hours minimum. The detonators, maintained at the conditioning temperature, shall be functioned as specified in 4.5.8.1. If any detonator fails to meet the requirements on the first attempt it shall be classed defective. If one or more defectives occur the lot shall be rejected.

4.5.10 Heat Seal Test of Sealed Bag.-The heat seal shall be verified as specified in MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level B.-Electric detonators shall be packaged in accordance with Code of Federal Regulations Title 49 Par 0-190. In addition, each detonator shall be electrically shorted during shipment and storage by means of a metallic connection between electrodes. When destined for shipment and uncovered storage in excess of 180 days from the loading date shown on the packing box to the date of the assembly into the end item, the following shall be included:

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Each intermediate carton, set-up box or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E of MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of MIL-P-116.

5.1.2 Level C.-Unless otherwise specified, the electric detonators shall be packaged in accordance with Paragraph 5.1.1 except that the barrier bag is not required.

5.2 Packing

5.2.1 Level B.-A suitable number of packaged detonators shall be packed in an outer box in accordance with Code of Federal Regulations 49 for Transportation of Explosives and Other Dangerous Articles, Paragraph 173.66.

5.2.2 Level C.-Unless otherwise specified, electric detonators shall be packed as specified in 5.2.1.

5.3 Marking.-Each inner container and intermediate carton set-up box or comparable container shall be printed, stamped or labeled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. (Number) Detonator, Electric, PA500
Lot No. Dated Packed (Mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to TT-I-1795.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with dwg. 8796522. The DOT designation shall be: Electric Blasting Caps - Handle Carefully (see NOTE).

e. NOTE: "The number of caps packed in one (1) outer pack shall be shown in the marking."

6. NOTES

6.1 Ordering Data.-See MIL-A-48078 (PA).

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6.2 Submission of Inspection Equipment Designs for Approval.- See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

6.3 Check Test for Possible Deterioration.-The following test is designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in that contract. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration. Two hundred (200) detonators shall be selected from each lot for this test.

6.3.1 Functioning.-The detonators shall be selected at random from each lot. The detonators shall be tested in the following sequence.

a. Measure resistance between electrodes, using a current limited to one milliampere. The resistance shall be no less than 2.5 ohms nor more than 4.0 ohms. If any detonator fails to meet the requirement, it shall be classed defective. If four or more defectives occur, the lot shall be rejected.

b. The detonators that are not classed defective in step a shall be functioned as specified in 4.5.8.1 and meet the requirements of 3.6.1. If any detonator fails to meet the requirements, it shall be classed defective. If one or more defectives occur, the lot shall be rejected.

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Preparing Activity:
Army-PA

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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